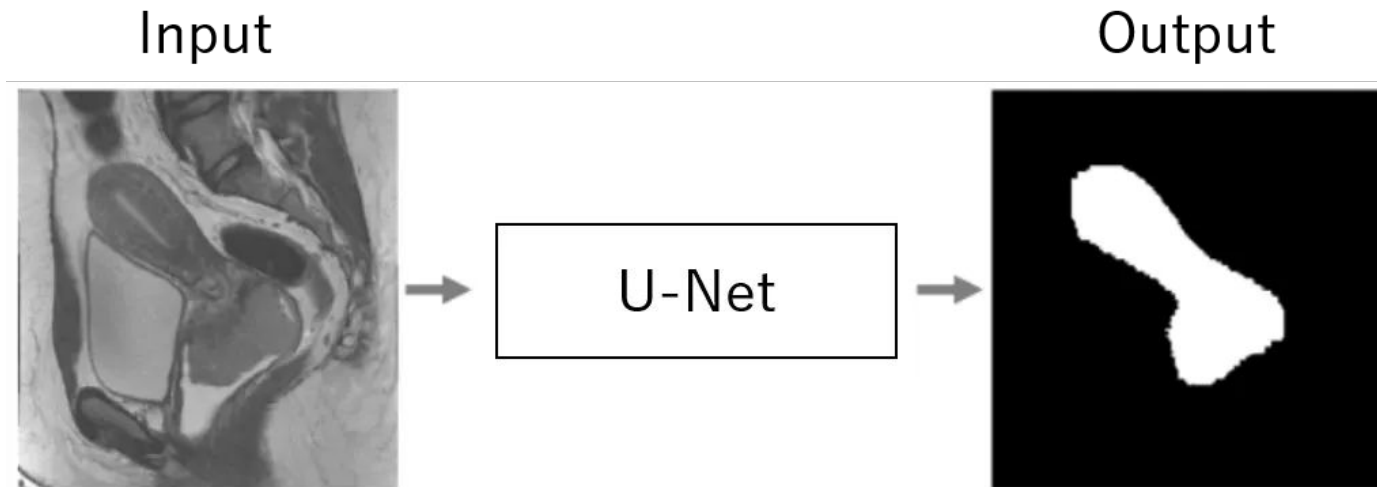


U-Net

Convolutional Networks for Biomedical Image Segmentation

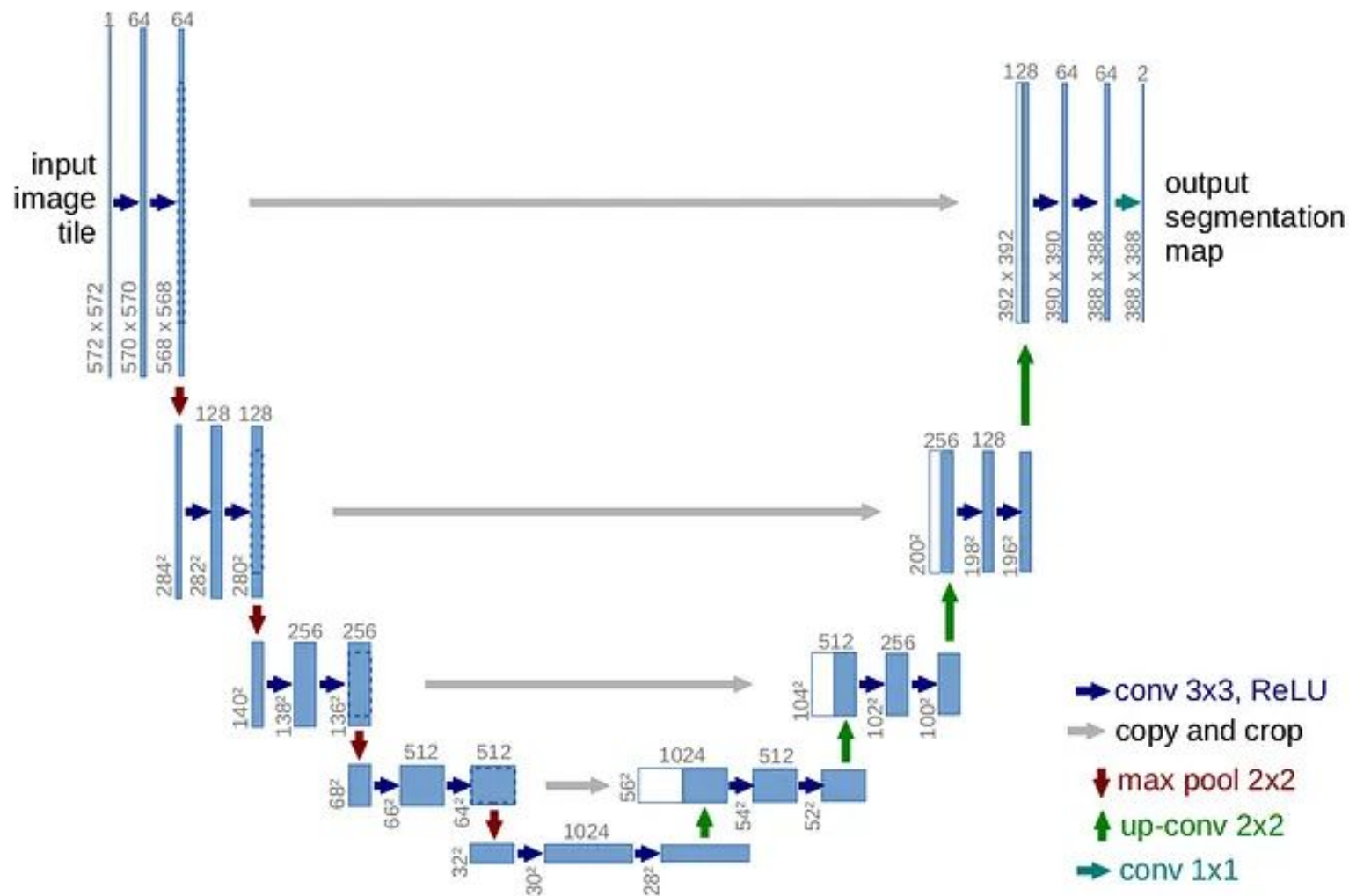
U-Net History

- CNN developed for biomedical image segmentation
- Architecture has a U-shaped structure
- Introduced in this [paper](#)
- Captures both context and fine details



U-Net Components

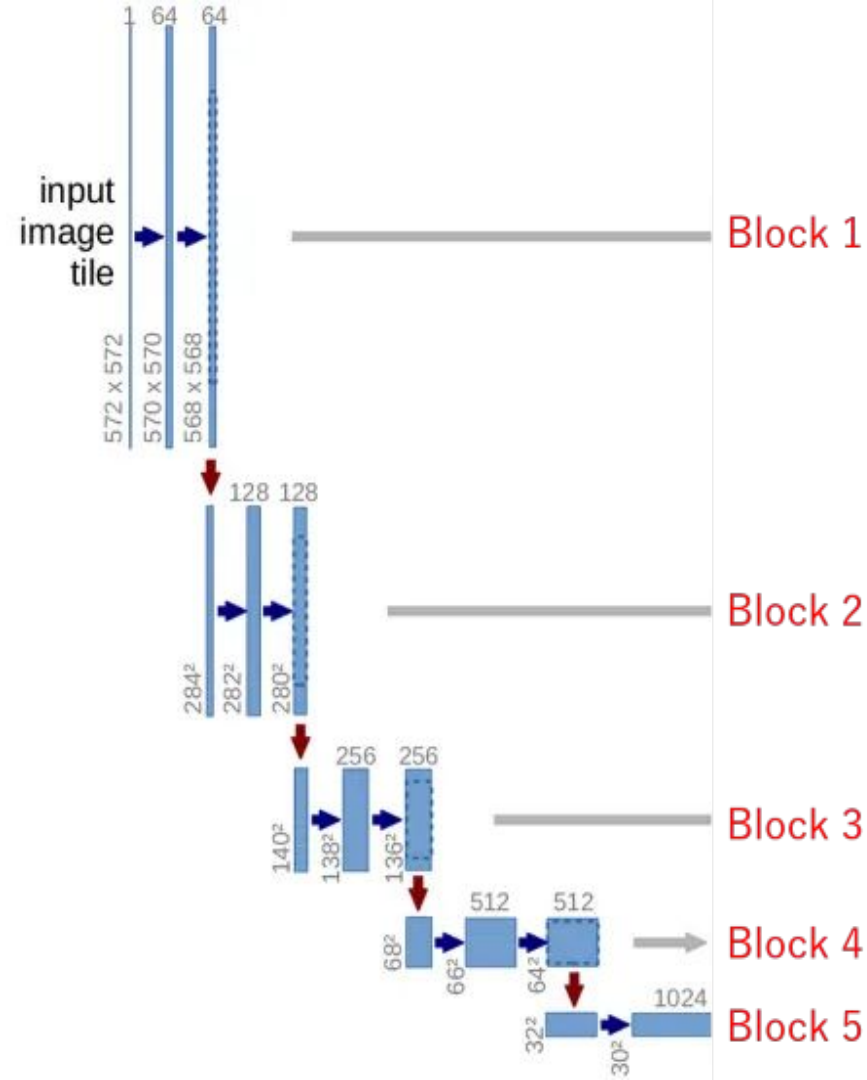
- Has two main components:
 - **Contracting Path:** Reduces the spatial dimensions of the image, while captures relevant information from the image.
 - **Expanding Path:** Upsamples the feature map and produces relevant segmentation map using patterns learnt in contracting path.
- Skip connections are used to connect corresponding encoder layers to decoder layers to **preserve spatial information**.



Contracting Path

- **Block 1:**

- An input image with dimensions 572^2 is fed into the U-Net. This input image consists of only **1 channel**, likely a grayscale channel.
- Two **3×3 convolution layers** (unpadded) are then applied to the input image, each followed by a **ReLU** layer. At the same time the number of channels are increased to 64 in order to capture higher level features.
- A 2×2 max pooling layer with a stride of 2 is then applied. This downsampled the feature map to half its size, 284^2 .



Contracting Path

- **Block 2:**

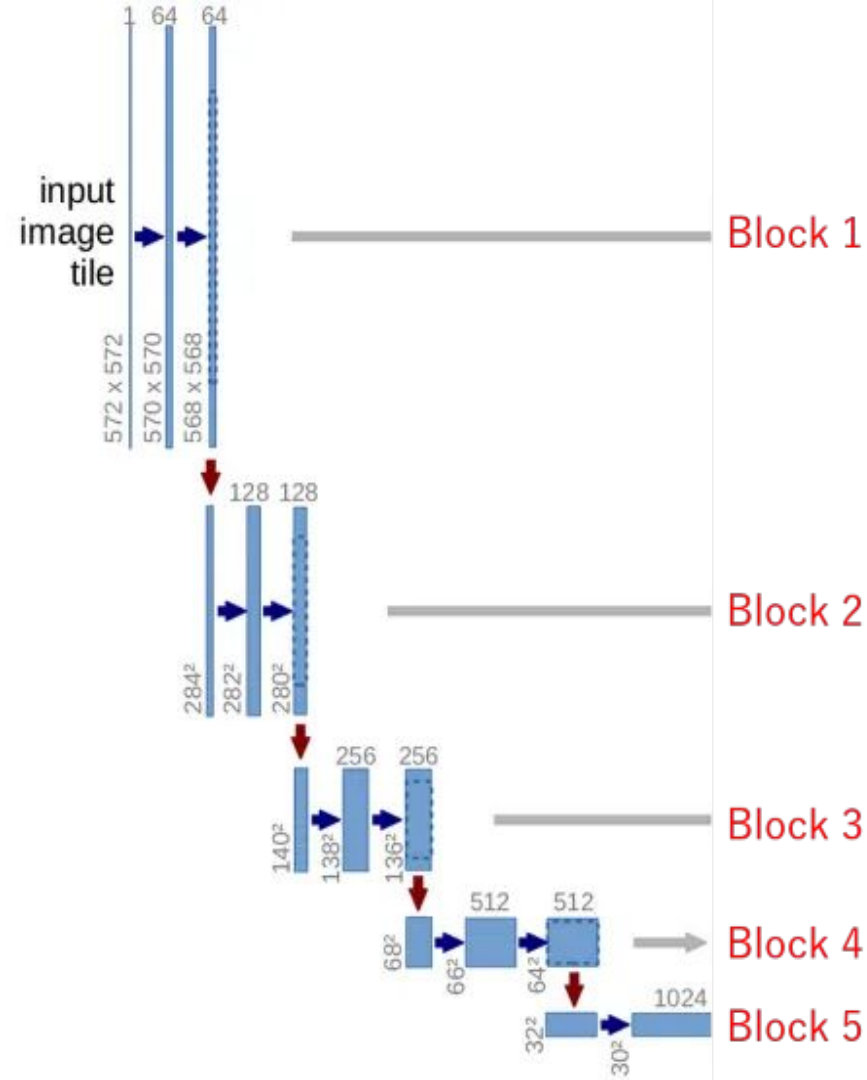
- Just like in block 1, two 3x3 convolution layers (unpadded) are applied to the output of block 1, each followed again by a ReLU layer. At each new block the number of feature channels are doubled, now to 128.
- Next a 2x2 max pooling layer is again applied to the resulting feature map reducing the spatial dimensions by half to 140^2 .

- **Block 3 & 4**

- The procedure used in block 1 and 2 is the same as in block 3 & 4.

- **Block 5**

- In the final block of the contracting path, the number of feature channels reach 1024 after being doubled at each block.
- This block also contains two 3x3 convolution layers (unpadded), which are each followed by a ReLU layer.



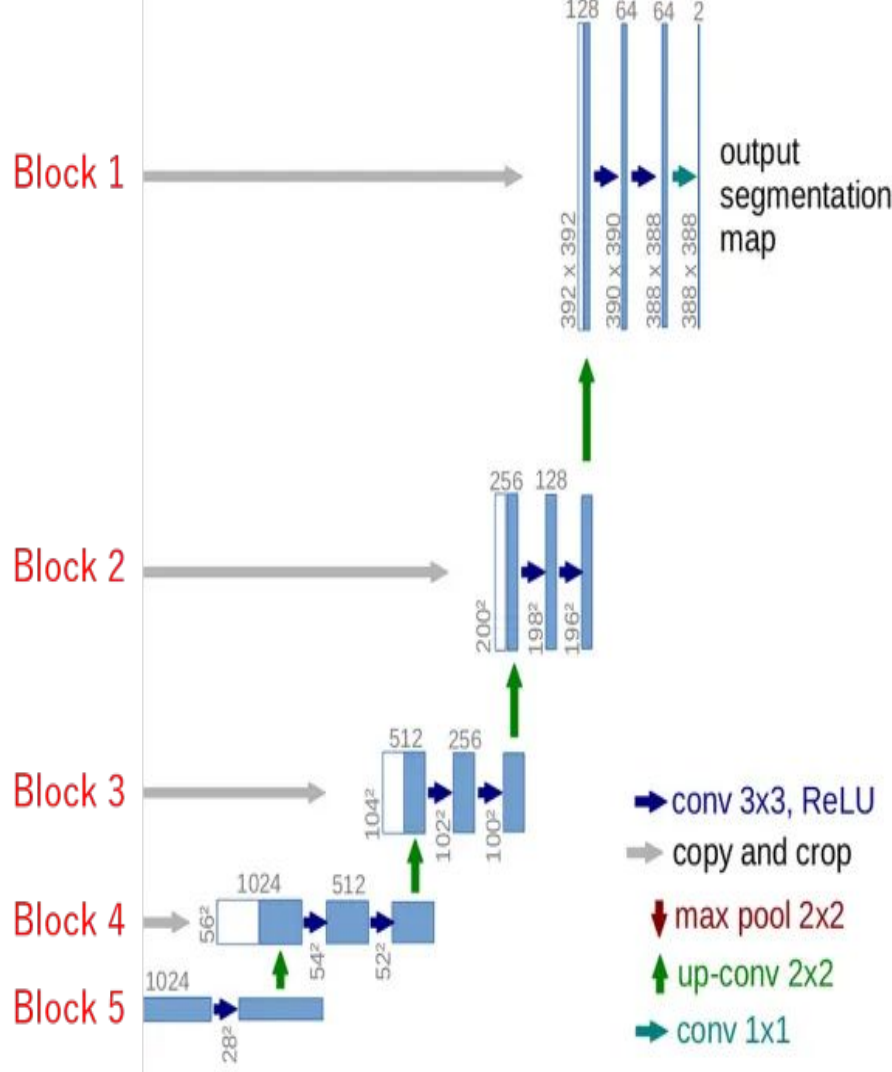
Expanding Path

- **Block 5:**

- Continuing on from the contracting path, a second 3x3 convolution (unpadded) is applied with a ReLU layer after it.
- Then a 2x2 convolution (up-convolution) layer is applied, upsampling the spatial dimensions twofold and also halving the number of channels to 512.

- **Block 4**

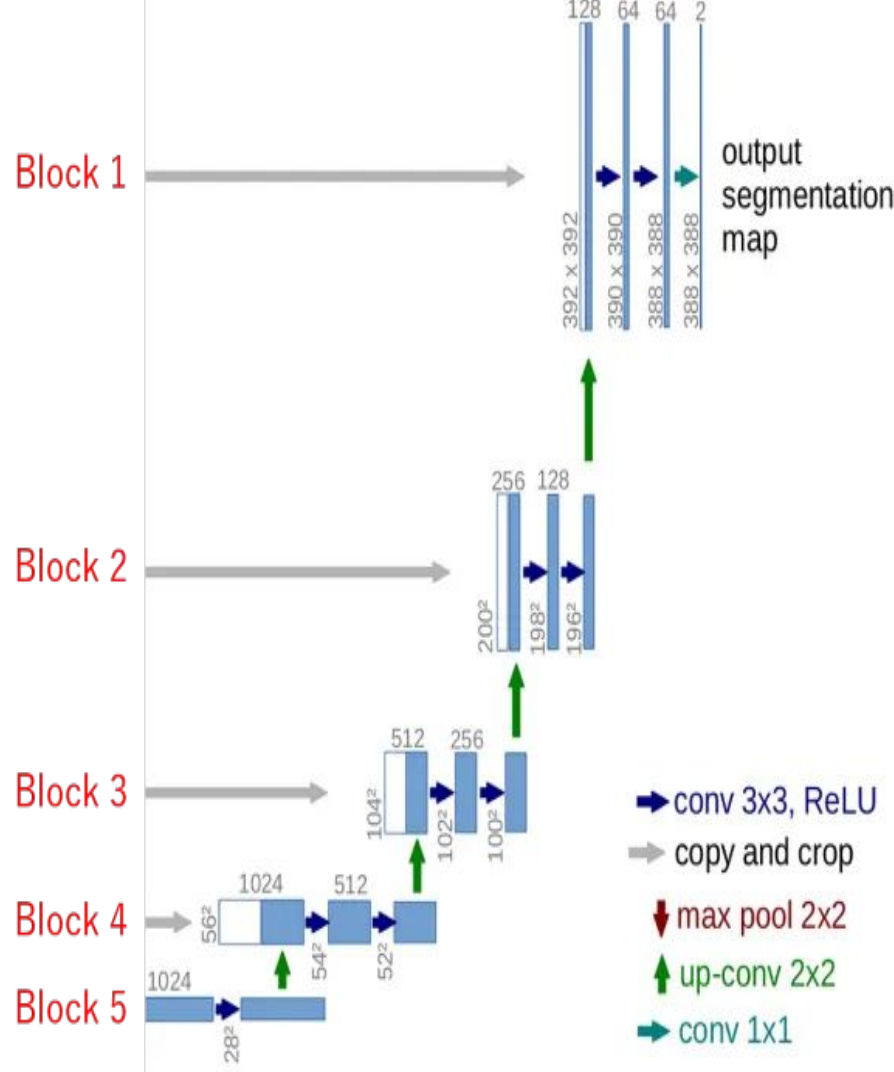
- Using skip connections, the corresponding feature map from the contracting path is then concatenated, doubling the feature channels to 1024. Note that this concatenation must be cropped to match the expanding path's dimensions.
- Two 3x3 convolution layers (unpadded) are applied, each with a ReLU layer following, reducing the channels to 512.
- After, a 2x2 convolution (up-convolution) layer is applied, upsampling the spatial dimensions twofold and also halving the number of channels to 256.



Expanding Path

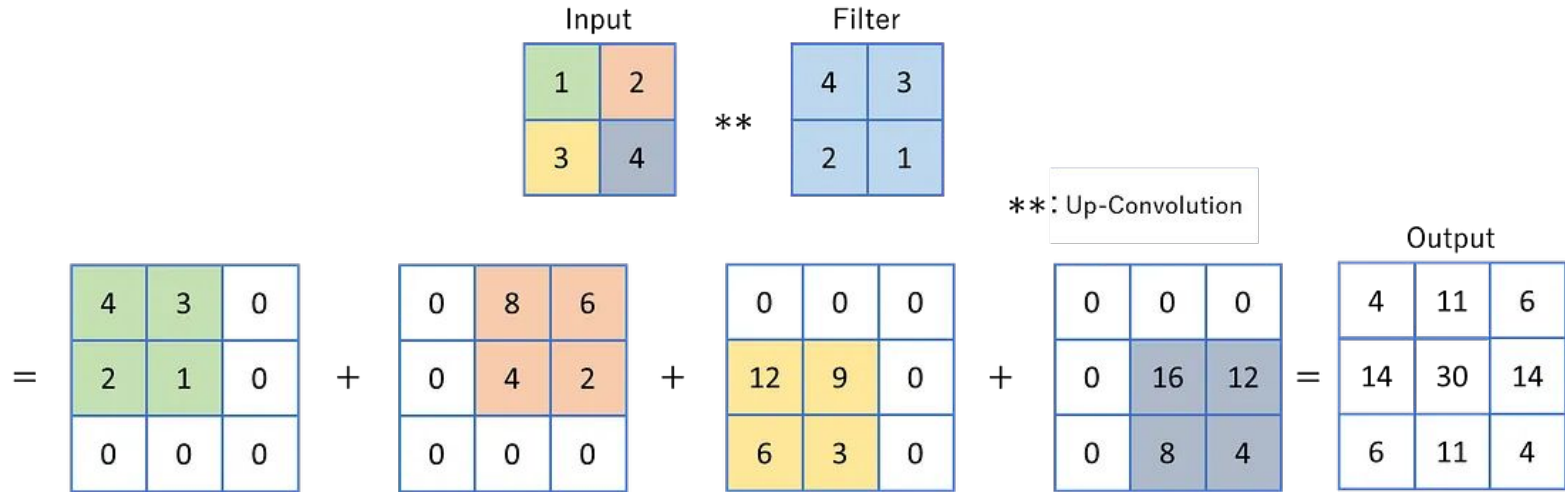
- **Block 3 & 2:**
 - Same as block 5 and 4.
- **Block 1**
 - In the final block of the expanding path, there are 128 channels after concatenating the skip connection.
 - Next, two 3x3 convolution layers (unpadded) are applied on the feature map, with ReLU layers in between reducing the number of feature channels to 64.
 - Finally, a 1x1 convolution layer, followed by an activation layer (sigmoid for binary classification) is used to reduce the number of channels to the desired number of classes.

After upsampling the feature map in the expanding path, a segmentation map should be generated, with each pixel classified individually.



Up-Convolution

- Known as deconvolution or transpose convolution
- Used to upsample images and recover spatial information



Skip Connections

- Used to send images directly from the contracting path to the expanding path
- Allows both high and low level features to be preserved and learnt

<https://medium.com/@alejandro.itoaramendia/decoding-the-u-net-a-complete-guide-810b1c6d56d8>

<https://medium.com/data-science/a-detailed-explanation-of-the-attention-u-net-b371a5590831>